CAPSTONE PROJECT II

PCA-Based Data-Driven Alloy Design and Analysis of Bulk Metallic Glasses (BMGs)

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1. Executive summary of the problem

The objective is to reduce a multivariable BMG dataset to interpretable principal components, identify compositional and descriptor trends, and shortlist alloys for validation. PCA is treated as an exploratory projection, not as a direct property-prediction or optimization model.

The supplied workbook contains 594 records, 11 complete numeric descriptors, and critical casting diameter Dmax. Composition labels exist for 410 records; 405 pass the stated formula-total audit and yield 38 elemental concentration features. These two feature spaces must be analyzed separately. [file:1]

Main result	Interpretation
Descriptor PCA: PC1 66.49%, PC2 12.70%; total 79.19%	Useful two-dimensional overview; four PCs retain 92.54%.
Composition PCA: PC1 10.14%, PC2 6.29%; total 16.44%	Two dimensions are inadequate for general compositional interpretation; 25 PCs retain at least 90%.
Data quality	184 composition labels are missing, five formulas fail the total check, and repeated compositions can have sharply conflicting Dmax values.
Prediction check	Grouped five-fold validation gives random-forest R2 = 0.091 and MAE = 2.960 mm; evidence is insufficient for reliable optimization of new alloys.

Five existing compositions are shortlisted as validation targets spanning high recorded Dmax and distinct score-space regions. Extreme records such as Pd77Cu6Si17 (2 and 72 mm) and Cu46Zr54/Zr54Cu46 (2 and 50 mm) are audit priorities, not accepted optima. No thermal-window or strength ranking is possible because those measurements are absent. All numerical findings in this report are calculated from the supplied workbook. [file:1]

2. Data (one page maximum)

Source: BMG_PCA.xlsx, Sheet1. One row is treated as one recorded alloy observation, not automatically as one unique composition or an independently replicated experiment. No experimental provenance, uncertainty, cooling history, casting protocol, or descriptor dictionary is supplied. [file:1]

Field / subset	Observed content / handling
All records	594 rows × 14 columns; retain original worksheet row as the unique audit key. Alloy_ID has three duplicate entries and is not a PCA input.
Alloy Modified	410 nonmissing labels, 184 missing (30.98%). Do not forward-fill or infer the absent compositions.
Numeric descriptors	CalRadR1, E1, Phi1, nWS1, B1, Tm1, Alpha1, Hm1, MHc1, Ro1, k1. All 11 complete; exact physical definitions and units are unverified.
Measured outcome	Dmax(mm): 594 complete values. Min 0.2; Q1 2.0; median 3.5; Q3 6.0; max 72.0; mean 5.163; sample SD 5.992 mm.
Descriptor analysis	594 × 11 standardized numeric predictor matrix. Dmax and identifiers excluded from PCA.
Composition analysis	405 × 38 standardized elemental matrix; 386 unique normalized compositions. Nonpresent elements are assigned zero, not treated as missing.
Composition convention	Formula numbers provisionally interpreted as atomic percentages. Accept sums within 100 ± 0.15 and renormalize accepted rows to exactly 100. Confirm the atomic/weight basis with the data owner.
Unavailable outcomes	No Tg, Tx, supercooled-liquid-region width, measured strength, hardness, or ductility. Do not reinterpret descriptor codes as measured alloy properties.

Five excluded formulas and worksheet rows: row 35, Cu58.8Hf19.6Ti14.7Nb2 (95.1); row 212, Pd81.5Cu2Si6.5 (90); row 224, Ti50Ni30Cu32Sn3 (115); row 298, Fe57.6Co9.6B19.2Si4.8Nb4 (95.2); row 364, Fe74.25B9.9Si9.9Zr1 (95.05). These are not silently corrected. The two accepted near-100 totals are 99.9 and 100.096. [file:1]

Family labels use the highest-concentration element after normalization; exact ties use alphabetical element order. These operational labels are not authoritative metallurgical family classifications. The workbook delivered with this report contains audit flags, standardized matrices, scores, coefficients, variance tables, predictions, and shortlist values. [file:1]

3. Explore the data

3.1 Distribution and representativeness

Figure 1 shows a right-skewed Dmax distribution: most records lie at small diameters, with a few very large observations. The mean exceeds the median. The labelled subset is dominated by Cu (78), Fe (68), Ca (52), La (47), Mg (40), and Zr (37) families; smaller families have much weaker statistical representation. [file:1]

Figure 1 – Dmax histogram and family boxplots

Insert the original figure here. This plot shows the distribution of individual Dmax records, which can only be drawn from the BMG_PCA.xlsx data (not supplied with the reference reports).

Figure 1. (a) Histogram of recorded Dmax for all 594 observations. (b) Boxplots for eight selected dominant-element families in the 405-row composition-audited subset. Outliers are retained; family differences may reflect sampling and processing as well as composition.

3.2 Duplicate-composition and provenance audit

Chemical identity was compared using normalized elemental vectors, so a different ordering of element symbols does not create a new alloy. The 405 accepted rows represent 386 unique compositions. Repeated rows are retained for exploration, because different outcomes might reflect different processing; their independence is not assumed. [file:1]

Equivalent composition	Recorded Dmax (mm)	Consequence
Pd77Cu6Si17	2 and 72	Same predictor position, radically different outcomes.
Cu46Zr54 = Zr54Cu46	2 and 50	Highest Zr binary record requires provenance verification.
Cu50Zr50	3 and 25	Cannot use maximum alone as a reproducible response.
La62Al14Cu12Ni12	3, 10, 10	Use range when selecting this composition.
Mg65Cu25Gd10	4 and 9	Replicate under controlled processing.

These conflicts cannot be adjudicated from the workbook. They might represent genuine process differences, transcription errors, or merged source records. No extreme value is removed solely because it is large. Unknown-composition records remain eligible for descriptor exploration, but not for a composition-based shortlist. [file:1]

3. Explore the data (continued)

3.3 Correlation structure

Figure 2 supports a strongly redundant descriptor block. nWS1 and B1 correlate at r = 0.953; Tm1 and Hm1 at 0.951; B1 and Hm1 at 0.942. Opposing trends include nWS1 and k1 (r = -0.813), B1 and MHc1 (-0.789), and E1 and MHc1 (-0.769). These are empirical descriptor associations, not established causal mechanisms or measured strength relationships. [file:1]

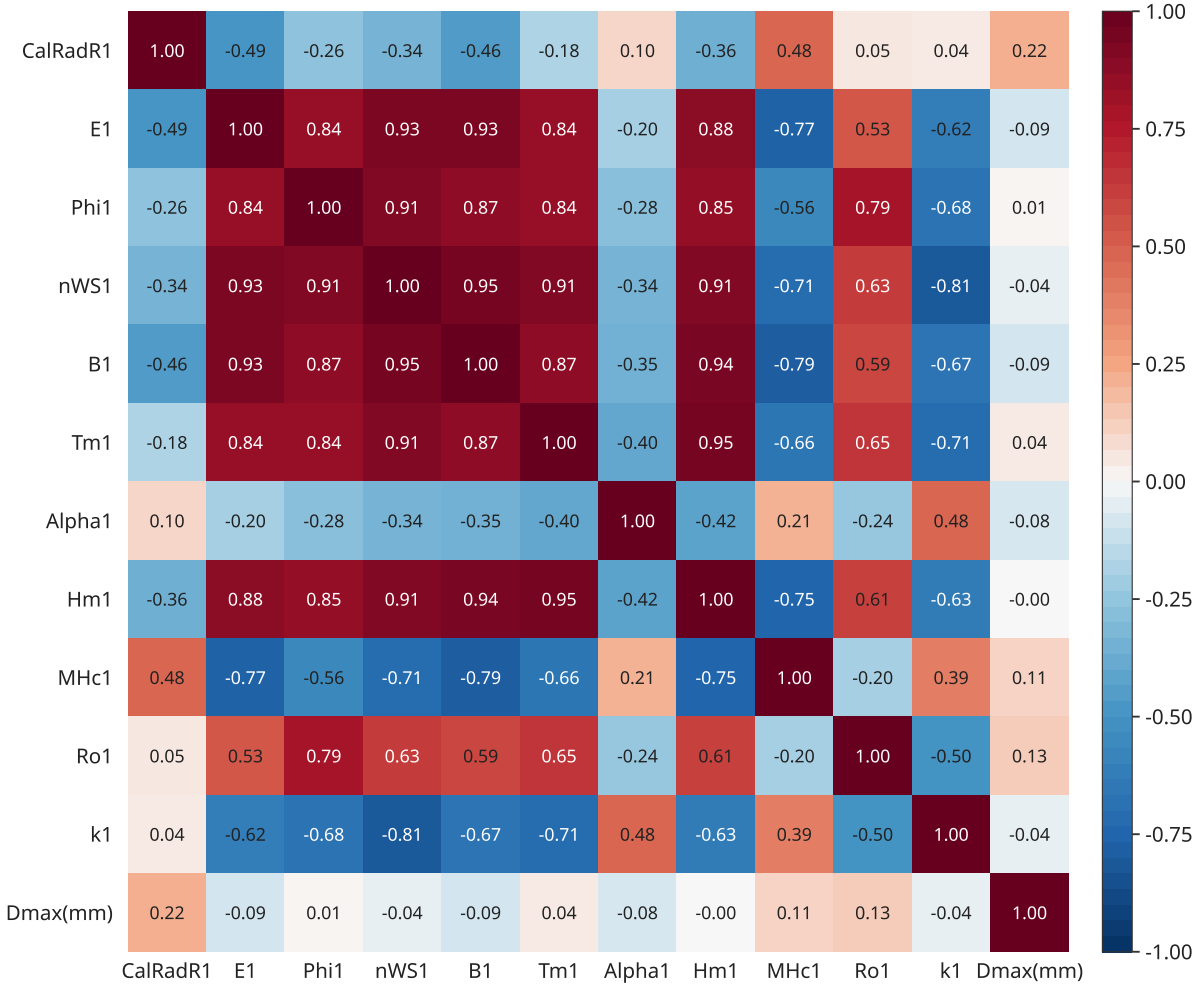


Figure 2. Pearson correlation matrix for the 11 numeric descriptors and Dmax, using all 594 rows. The coded column names are preserved because their physical definitions and units have not been verified.

No individual descriptor has a strong linear association with Dmax: the largest absolute correlation is CalRadR1, r = 0.222; Ro1 is 0.126 and MHc1 is 0.107. Among the 405 accepted compositions, Ce (0.151), Pd (0.147), Al (0.133), and Zr (0.133) have weak positive concentration correlations; Fe (-0.143), Ni (-0.132), and B (-0.131) have weak negative correlations. These pooled associations are sensitive to family sampling, rare elements, closure, and extreme outcomes; they do not justify increasing or decreasing an element in an alloy design. [file:1]

4. Methodology

4.1 Cleaning and feature-space separation

Whitespace was removed from labelled formulas, element-symbol/number tokens were parsed, repeated symbols were summed, and totals were audited. Missing formulas were not imputed. Accepted concentration vectors were renormalized to sum to 100. Five severely nonclosing formulas were excluded only from composition-dependent analyses, not from the complete-descriptor PCA. The numerical tolerance is an explicit analysis choice, not proof of chemical correctness. [file:1]

Two independent PCAs were fitted: (A) the 11 numeric descriptors for 594 records and (B) the 38 elemental concentrations for 405 accepted records. The matrices were not concatenated: doing so would mix feature representations and double-count potentially related compositional information. Dmax was held out of both decompositions and used only as a supplementary color/outcome.

4.2 Standardization and PCA equations

For each feature j, z_ij = (x_ij - mean_j)/s_j. StandardScaler uses the population standard deviation (ddof = 0). This gives all nonconstant columns equal input variance despite their different scales. Standardization is especially consequential for rare elements, which otherwise have very small raw concentration variance. [web:5]

For standardized Z, form S = Z^T Z/(n - 1). The eigenvectors v_k define PC directions; scores t_k = Z v_k locate the alloys. The eigenvalue lambda_k is the score variance. Explained-variance ratio = lambda_k / sum(lambda), and cumulative variance is the sum over retained PCs. PCA seeks variance, not covariance with the response. [web:10]

The loading heatmaps report eigenvector coefficients v_jk, matching scikit-learn components_. Biplot arrows instead use correlations corr(z_j, t_k), multiplied by a common display factor of 3. Scores retain their raw PC coordinates. Arrow and point lengths therefore have different units; neither arrow length nor a point-arrow distance is a predicted Dmax. PC signs are arbitrary; relative directions and absolute coefficients carry the interpretation.

4.3 Validation design

Exploratory PCA uses the available full matrix for descriptive purposes only. For predictive evaluation, restrict to the 405 composition-audited records and hold all records of the same normalized composition in the same fold. Use five-fold GroupKFold. Fit scaling and PCA inside each training-fold pipeline, then transform the held-out fold; fitting them globally before predictive validation would leak information. [web:3]

Compute out-of-fold MAE, RMSE, and pooled R2 on raw Dmax in millimetres. Hyperparameters are fixed in advance for illustrative benchmarking rather than optimized on the test folds. This validation tests held-out compositions within the sampled dataset, not transfer to wholly unseen alloy families or casting protocols.

4. Methodology and PCA results

4.4 Scree, explained variance, and dimensional adequacy

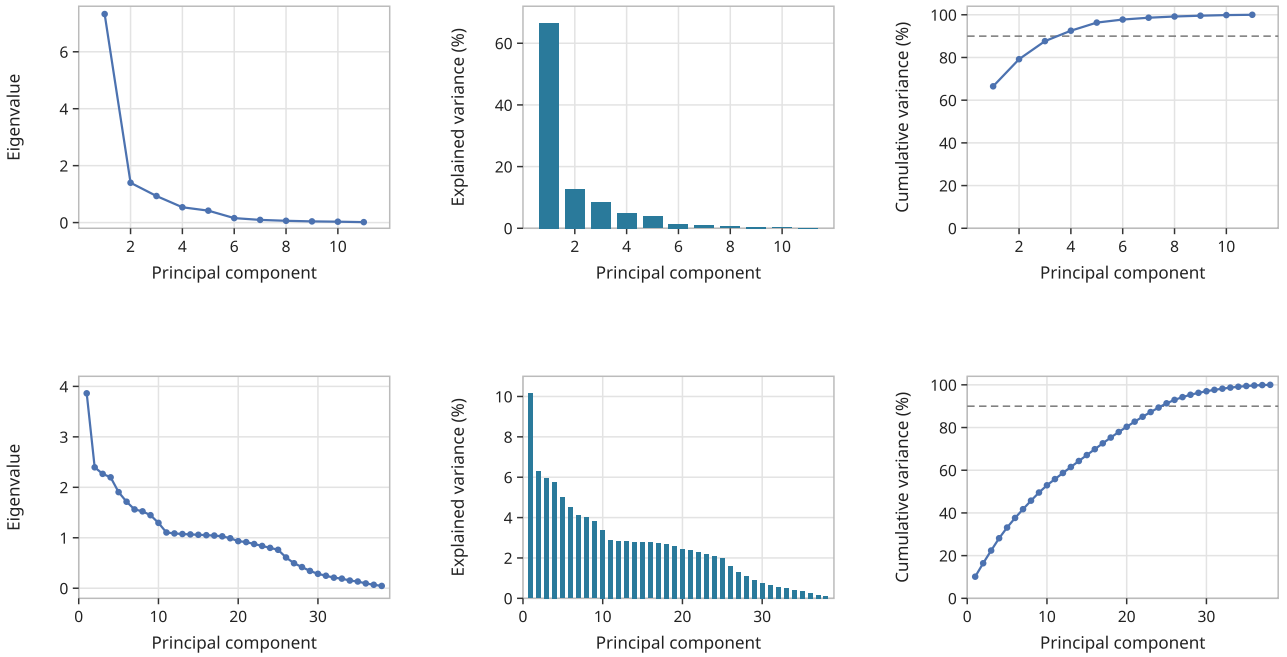


Figure 3. Scree plots, individual explained variance, and cumulative variance for descriptor PCA (top) and elemental-composition PCA (bottom). The dashed cumulative line marks 90%; scales differ because the matrices contain 11 and 38 features, respectively.

Feature space	PC1	PC2	PC1 + PC2	At least 90%
11 descriptors; n = 594	66.49%	12.70%	79.19%	4 PCs; 92.54%
38 elements; n = 405	10.14%	6.29%	16.44%	25 PCs

The descriptor scree plot drops sharply after PC1, followed by smaller contributions. PC3 adds 8.47% (87.65% cumulative); PC4 adds 4.88%. Two dimensions are useful for a broad descriptor overview but omit 20.81% and cannot support every local or property-specific comparison. Four or five PCs are preferable for higher-fidelity distance-based analysis; five retain 96.34%. [file:1]

The composition scree plot has no comparably dominant first component. The first four PCs together retain only 28.15%; 20 PCs are needed for 80.35%, 25 for at least 90%, and 28 for at least 95%. Consequently the elemental PC1-PC2 plane is not an adequate summary of the whole chemical search space. This is not inconsistent with the descriptor result: the two matrices represent different variables and different row subsets. [file:1]

All quoted percentages refer to variance in the specified standardized predictor matrix, not explained variation in Dmax, prediction accuracy, or the probability that a candidate will form glass.

4. PCA results: loadings

Figure 4 identifies the variables contributing most strongly to each axis. Values below are eigenvector coefficients; domination means large absolute coefficients, not causality. [file:1]

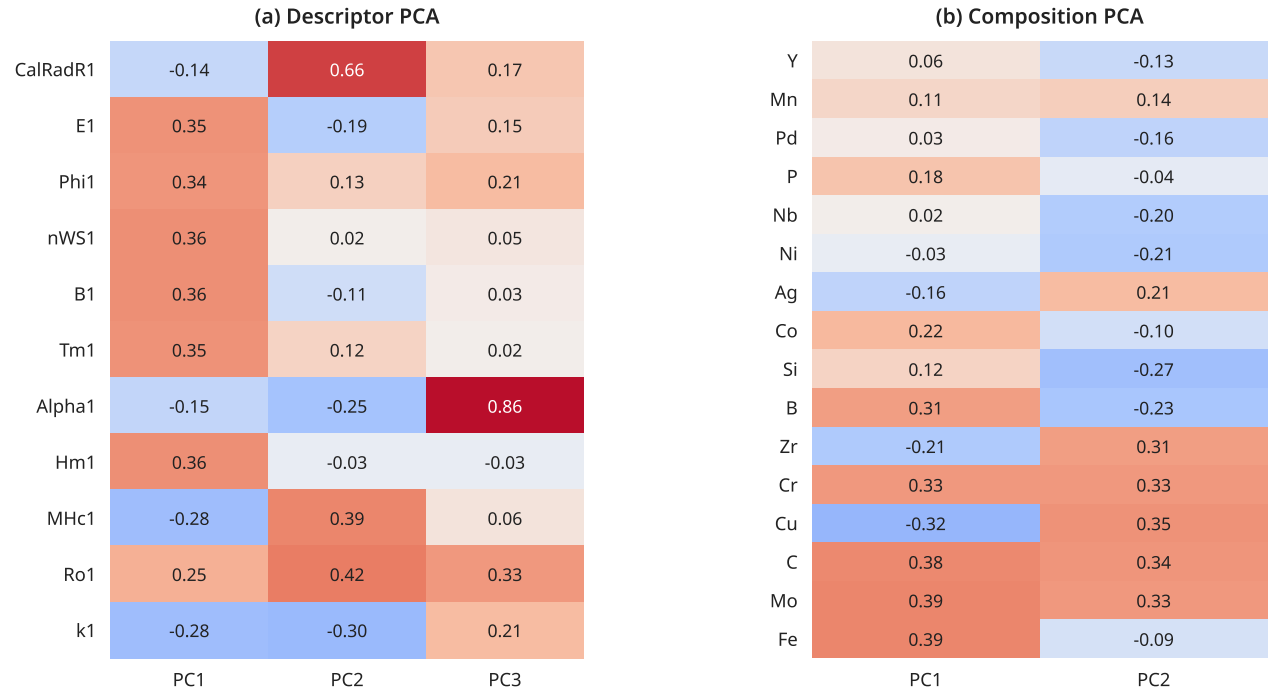


Figure 4. Loading-coefficient heatmaps. (a) All descriptor coefficients for PC1-PC3. (b) Sixteen elements selected by their maximum absolute coefficient on composition PC1 or PC2. Full coefficient matrices are supplied in the analysis workbook.

4.5 Which concentrations dominate each principal component?

Composition PC1: strongest positive coefficients are Fe (+0.389), Mo (+0.385), C (+0.384), Cr (+0.329), and B (+0.306). Strongest negative coefficients are Cu (-0.317), Zr (-0.210), and Ag (-0.157). This axis contrasts portions of the Fe/metalloid-rich population with Cu/Zr-rich compositions. [file:1]

Composition PC2: positive Cu (+0.353), C (+0.337), Cr (+0.334), Mo (+0.327), and Zr (+0.310) oppose Si (-0.268), B (-0.229), Ni (-0.205), and Nb (-0.197). These first two axes account for little total elemental variance, so an element with a small coefficient here may still matter on later PCs or within a particular family. [file:1]

4.6 What do the descriptor axes represent?

Descriptor PC1 has large positive coefficients for nWS1 (+0.363), B1 (+0.359), Hm1 (+0.355), Tm1 (+0.346), E1 (+0.345), and Phi1 (+0.340); k1 (-0.279) and MHc1 (-0.277) point oppositely. PC2 is dominated by CalRadR1 (+0.663), Ro1 (+0.417), and MHc1 (+0.391), opposed by k1 (-0.305) and Alpha1 (-0.246). Alpha1 dominates PC3 (+0.858), showing one important contrast lost in a two-dimensional view. [file:1]

4. PCA results: scores and alloy families

Figure 5 shows partial family organization, not clean universal clusters. In descriptor space, Ca-rich alloys lie mainly at negative PC1/negative PC2, Mg-rich alloys occupy a nearby but less negative PC1 region, and La-rich alloys tend toward negative PC1/positive PC2. Fe-rich alloys generally lie at positive PC1; Cu/Zr groups overlap around central PC1 and positive PC2. [file:1]

Figure 5 – Descriptor and composition score plots

Insert the original figure here. This plot shows the individual PCA scores of each alloy record, which can only be drawn from the BMG_PCA.xlsx data (not supplied with the reference reports).

Figure 5. PC1-PC2 score plots colored by operational dominant-element family. (a) The 405 composition-audited records projected onto the descriptor PCA fitted to all 594 records. (b) The same 405 records projected onto composition PCA. Nine common families are shown individually; the rest are pooled as Other for display only.

Family	Median descriptor PC1	Median descriptor PC2
Ca	-4.52	-1.38
Mg	-2.34	-1.02
La	-2.56	+1.87
Fe	+3.73	-1.01
Cu	+0.60	+0.67
Zr	+0.38	+1.02

Using all 16 unpooled operational family labels, the two-dimensional silhouette score is 0.095 for descriptor scores and -0.145 for composition scores. This internal geometric diagnostic supports substantial overlap rather than strongly separated family groups. It is not a clustering accuracy score and is sensitive to unequal family sizes and the family-label convention. [file:1]

The composition score plot can highlight some Fe-versus-Cu/Zr contrasts but compresses many other families together because 83.56% of elemental variance lies outside this plane. Any subsequent clustering should therefore be performed and stability-checked in a higher-dimensional representation, not solely on this plot. [file:1]

4. PCA results: biplot interpretation

Figure 6 – Supplementary-outcome biplots

Insert the original figure here. This plot shows the individual PCA scores of each alloy record, which can only be drawn from the BMG_PCA.xlsx data (not supplied with the reference reports).

Figure 6. Supplementary-outcome biplots. Points are colored by log10(recorded Dmax/mm); Dmax is not a PCA input or loading arrow. Arrows are variable-score correlations multiplied by 3 for display; left shows all descriptors, right the ten elements with largest PC1-PC2 correlation-vector norms. Circled labels 1-5 identify the shortlist in Section 6. Coincident compositions may have different outcome colors.

In the descriptor biplot, aligned descriptor arrows correspond to the correlated block already seen in Figure 2. Positive PC2 points are relatively associated with CalRadR1, Ro1, and MHc1, whereas negative PC2 favors the opposite pattern. Nevertheless, descriptor PC1 correlates with Dmax at only -0.026 and PC2 at +0.260. Composition score correlations with Dmax are -0.049 and +0.107. A high-variance axis is therefore not a reliable high-performance direction. [file:1]

Shortlist alloys 1 and 2 occupy similar positive-PC2 descriptor and Cu/Zr-associated composition regions. Alloy 3 lies on the opposite side of descriptor PC2 and on positive composition PC1, providing a chemically different high-recorded-diameter comparison. Alloys 4 and 5 sample negative descriptor-PC1 regions, with opposite signs of PC2, giving a broader validation portfolio. The composition map alone cannot certify their separation because it retains only 16.44% of variance. [file:1]

4.7 What cannot be inferred from these plots?

A point lying near an arrow is not proof that adding the associated element increases glass-forming ability. Arrow angles approximate correlations only in the retained projection; this is particularly weak for the low-variance composition plane. Dmax colors are observed supplementary information, not PCA predictions.

The supercooled liquid region is Delta Tx = Tx - Tg, involving the crystallization onset and glass-transition temperatures. Neither temperature is available here; no wide-Delta-Tx composition can be identified. [web:2]

The workbook also lacks measured mechanical strength, so no alloy can be ranked for high strength from these data. Descriptor names such as E1 or B1 must not be substituted for experimentally measured alloy strength. [file:1]

5. Prediction model

PCA itself does not predict Dmax. To test whether an additional supervised model offers useful evidence, five fixed models were evaluated on the 405 composition-audited records using composition-grouped five-fold out-of-fold predictions. The inputs are the 11 descriptors, not Dmax or identifiers. No missing-formula rows enter this benchmark. [file:1]

Model	MAE (mm)	RMSE (mm)	Pooled R2
Mean baseline	3.436	5.933	-0.006
PCR (2 PCs)	3.379	5.857	0.020
PCR (4 PCs)	3.394	5.936	-0.007
Ridge (11 descriptors)	3.400	5.961	-0.016
Random forest	2.960	5.639	0.091

PCR uses standardized descriptors, either two or four training-fold PCs, and Ridge(alpha = 10). Full-descriptor ridge also uses alpha = 10. The random forest uses 200 trees, min_samples_leaf = 5, max_features = 0.8, and seed 42. The baseline predicts each training fold's mean. Parameters were not tuned; results are an illustrative comparison rather than a definitive model-selection study.

Figure 7 – Recorded vs. predicted Dmax

Insert the original figure here. This plot shows the out-of-fold prediction for each alloy record, which can only be drawn from the BMG_PCA.xlsx data (not supplied with the reference reports).

Figure 7. Recorded versus grouped out-of-fold random-forest predictions. The dashed line indicates ideal agreement. Predictions fail to reproduce the largest observed diameters, consistent with low pooled R2.

The forest has the smallest MAE (2.960 mm) and RMSE (5.639 mm), but R2 = 0.091 means poor predictive coverage overall. Two-PC regression has R2 = 0.020; retaining high predictor variance does not guarantee good prediction. The conflicting outcomes at identical compositions place an additional limit on composition-only predictability. [file:1]

Before using a model to propose new alloys: recover provenance, reconcile duplicates, add process variables, compare raw/log-target models, tune within nested grouped validation, and test uncertainty calibration. Family-held-out and external validation are needed for claims of transfer beyond familiar chemistry. Scaling and PCA must remain inside the training pipeline. [web:3]

6. Optimization of solution

6.1 Evidence-based shortlist, not a universal optimum

The practical optimization target is a small, informative validation portfolio, not maximization of a PC score. Selection combines observed Dmax, composition validity, coverage of distinct descriptor-score regions, and explicit handling of repeat-record conflicts. The candidates below are existing dataset compositions proposed for further investigation; they are not newly discovered or experimentally validated designs. Concentrations follow the provisional formula-percentage convention. [file:1]

Label / composition	Original Dmax, mm	Descriptor PC1 / PC2	Worksheet rows
1. Zr50Cu48Ag2	30	+0.539 / +1.227	291
2. Zr48Cu34Pd2Ag8Al8	25	+0.216 / +1.115	290
3. Pd40Cu30Ni10P20	25	+0.803 / -1.106	217
4. La62Al14Cu12Ni12	3, 10, 10	-2.318 / +1.942	180, 197, 395
5. Mg65Cu25Gd10	9, 4	-2.404 / -1.068	143, 166

Score coordinates above use the descriptor PCA fit on all 594 records. For alloys 4 and 5 they refer to representative rows 197 and 143, respectively. The analysis workbook also provides composition-PC coordinates. All recorded repeat outcomes are retained in the table; single-record outcomes are not assumed reproducible. [file:1]

6.2 Why these five compositions?

1. Zr50Cu48Ag2: the 30 mm record is a high observed diameter in a formula-valid Cu/Zr/Ag region. Its positive descriptor PC2 is consistent with the CalRadR1/Ro1/MHc1 direction, while its negative composition PC1 and positive PC2 associate with Cu/Zr chemistry. Prioritize replication and local Ag-substitution tests. [file:1]

2. Zr48Cu34Pd2Ag8Al8: the 25 mm observation provides a multicomponent comparator in nearly the same plotted region as candidate 1. Their similar two-dimensional scores but different chemistries make them a useful check on whether the projection obscures relevant higher-dimensional differences. Test local substitutions only after establishing a reproducible baseline. [file:1]

3. Pd40Cu30Ni10P20: the 25 mm observation samples a different chemistry and negative descriptor PC2. It is a deliberate counterexample to choosing only positive-PC2 alloys and tests whether high Dmax occurs through multiple compositional pathways. Its economic suitability is not established by this dataset. [file:1]

4. La62Al14Cu12Ni12: the repeated 3-10 mm range and negative-PC1/positive-PC2 location make this a process-sensitivity target rather than a top-diameter claim. Reconcile the three source records and repeat under a fixed protocol. [file:1]

5. Mg65Cu25Gd10: the 4-9 mm range covers a negative-PC1/negative-PC2 region distinct from La-rich candidates. It broadens chemical coverage and supports a controlled repeatability study. No strength or density ranking is claimed from the supplied measurements. [file:1]

6. Optimization and validation plan

6.3 Separate audit controls from development candidates

Before accepting any record as the best glass former, audit Pd77Cu6Si17 (2 versus 72 mm) and Cu46Zr54/Zr54Cu46 (2 versus 50 mm). They occupy identical predictor coordinates within each composition pair, so PCA cannot resolve the disagreement. These two compositions should be rechecked as data-quality controls, not promoted based on the larger value alone. [file:1]

6.4 Targeted follow-up

Stage	Action	Decision criterion
Provenance and repeatability	Recover original publications, composition basis, casting geometry, fluxing/purity, thermal history, and measurement definitions; independently reproduce candidate baselines.	Accept Dmax only when the casting protocol and amorphous verification are documented.
Targeted casting	Cast stepped rod diameters around each reproduced baseline under controlled processing, using independent melts.	Bracket the largest reproducibly fully amorphous cross-section; quantify between-melt variability.
Structure and thermal tests	Use X-ray diffraction to assess amorphous structure and DSC to determine Tg and Tx at a stated heating rate; investigate ambiguous samples more closely.	Report structure and Delta Tx together rather than inferring either from PCA.
Mechanical testing	Measure compressive/tensile response or hardness under a specified geometry, strain rate, and replicated protocol.	Report strength, plastic strain, and scatter; do not infer them from Dmax alone.
Regression and clustering	Add measured process/thermal variables; evaluate regularized and nonlinear regression under grouped validation. Cluster in retained higher-dimensional scores and check stability.	Require improvement over a simple baseline, external validation, and calibrated uncertainty before model-guided ranking.
Local composition design	Within a verified alloy family, vary one or two substitutions while preserving a total of 100; compare to the parent composition.	Advance only improvements exceeding experimental uncertainty; retain composition and process constraints.

The casting and thermal characterization plan directly addresses the missing information: critical casting dimension is connected to cooling requirements, while the supercooled liquid region requires measured thermal transitions. These are distinct measurements, not interchangeable PCA outputs. [web:2]

After validation, a multiobjective design could seek high reproducible Dmax, adequate Delta Tx, required mechanical performance, and acceptable cost/safety. At present only recorded Dmax is available, so a defensible multiobjective optimum cannot be computed. Proposed numerical composition changes would be hypotheses requiring testing, not predicted improvements.

7. Conclusion statement

PCA successfully narrows the BMG search space by revealing strong redundancy among the 11 numeric descriptors and partial separation of several alloy families. Descriptor PC1 and PC2 retain 79.19% of standardized descriptor variability, whereas elemental PC1 and PC2 retain only 16.44%. Thus a two-dimensional descriptor map is useful for exploratory overview, but a two-dimensional elemental map is inadequate for the full compositional design problem. [file:1]

The dominant elemental contrasts are Fe/Mo/C/Cr/B versus Cu/Zr/Ag on composition PC1, with a second mixed Cu/C/Cr/Mo/Zr versus Si/B/Ni/Nb contrast on PC2. These are dataset-level patterns, not element-addition rules. Strong descriptor correlations coexist with weak Dmax associations, and the supervised benchmark also has limited predictive power. [file:1]

The recommended outcome is a five-composition validation portfolio: Zr50Cu48Ag2, Zr48Cu34Pd2Ag8Al8, Pd40Cu30Ni10P20, La62Al14Cu12Ni12, and Mg65Cu25Gd10. Their selection combines recorded diameter, contrasting score-space positions, and explicit repeat-record ranges. Extreme conflicting compositions are retained as audit controls, not advertised as optimum BMGs. [file:1]

Limitations and scope

The main limitations are missing composition labels; unverified descriptor definitions and units; inconsistent formula totals and repeated outcomes; absent process metadata; unbalanced families; rare-element upweighting under standardization; and compositional closure, which can induce correlations because concentrations sum to a constant. PCA is linear, sensitive to feature scaling and outliers, and unsupervised. It does not establish causality, model crystallization kinetics, or guarantee extrapolation to untested compositions.

Centered/log-ratio approaches merit a sensitivity analysis for compositional geometry, but many elements are truly absent, so zero handling must be physically justified rather than achieved by arbitrary pseudocounts. Other useful checks include covariance-PCA, robust scaling/PCA, bootstrap subspace stability, and within-family analysis. These are proposed follow-ups and were not performed in the present report.

Additional experiments must establish actual glass formation, casting-size reproducibility, thermal stability, and strength. Additional supervised analysis must use leakage-safe grouped validation and external experiments before any model-guided optimum is claimed. [web:2][web:3]

Traceability of results

All dataset-specific values derive from the uploaded BMG_PCA.xlsx [file:1]. Public methodological support: StandardScaler documentation [web:5], scikit-learn PCA/decomposition documentation [web:10], leakage-safe preprocessing guidance [web:3], and the study on supercooled-liquid behavior and critical casting dimensions [web:2]. The accompanying results workbook preserves worksheet row numbers and the exact outputs used in the tables and figures.

08 | KEY FINDINGS

8. Executive summary

This study uses Principal Component Analysis (PCA) to explore how 594 bulk metallic glass (BMG) alloy records are related through 11 composition-derived descriptors. The maximum critical casting diameter (Dmax) is kept outside the PCA calculation and is compared with the PCA results afterwards. This helps show patterns in the alloy descriptors without allowing Dmax to define those patterns.

The analysis shows that a small number of combined trends describe much of the variation in the dataset. The first two principal components explain 79.2% of the total variation, while the first three explain 87.7%. Alloy families occupy recognisable regions, and high Dmax values are more common in the Zr-Cu and La-rich regions. However, PCA does not directly predict glass-forming ability, strength or the supercooled liquid region.

Assignment question	Main finding
Q1. Variance in PC1 and PC2	PC1 explains 66.5% and PC2 explains 12.7%; together they explain 79.2%. This is useful for a first 2-D map, but four PCs are needed to pass 90%.
Q2. Dominant variables	PC1 is mainly associated with nWS, B, Hm, Tm, E and Phi. PC2 is led by CalRad, while Alpha strongly contributes to PC3. Element-level associations are discussed in Section 11.4.
Q3. Properties that vary together	nWS, B, Hm, Tm, E and Phi rise together. Thermal conductivity (k) and molar heat capacity (MHc) tend to move in the opposite direction; Alpha also partly opposes them.
Q4. Alloy-family grouping	The score plot shows recognisable regions for Ca-, Mg-, rare-earth-, Zr/Cu-, Pd-, Fe- and Co-rich alloys. A 5-nearest-neighbour check reports 84% cross-validated family recovery; Zr- and Cu-rich alloys overlap.
Q5. Dmax, Delta Tx and strength	High Dmax values are concentrated mainly in the Zr-Cu region and the La-rich region. The dataset has no Delta Tx or direct strength measurements, so these properties cannot be identified directly.
Q6. Candidate compositions	Seven compositions (C1-C7) are retained as candidates: high-Dmax anchors, lower-Dmax neighbours for follow-up, a La-based option and one Fe-based stiffness-oriented outlier.
Q7. PCA limitations	PCA is linear and unsupervised, the dataset is incomplete at the composition level, and Dmax is noisy. Regression, clustering, thermodynamic screening and experimental casting/testing are recommended next steps.

MAIN TAKEAWAY

PCA is useful for reducing the search space and identifying composition families. It should be treated as a screening method that generates hypotheses, not as proof that a particular alloy will form a glass or have high strength.

9. Background and project objective

Bulk metallic glasses are metallic materials in which atoms do not form the regular, repeating crystal arrangement found in conventional crystalline alloys. When a suitable liquid alloy is cooled quickly enough, crystallisation can be avoided and the disordered liquid structure can be retained as a solid glass. This structure can give BMGs high strength and a large elastic strain range; corrosion resistance is also good in some alloy systems.

The ability to form a glass depends on both composition and processing. Small changes in the elements present can affect atomic packing, chemical interactions and the tendency to form competing crystalline phases. Cooling rate, melt cleanliness and casting conditions also matter. For this reason, selecting a new BMG composition is a multi-variable materials-design problem.

In this project, PCA is used to reduce the 11 numerical descriptors to a smaller set of combined axes. The resulting plots are used to explore relationships between alloy records, identify broad alloy-family regions and shortlist compositions for future study. Dmax is the maximum critical casting diameter reported for an alloy; a larger value generally indicates that a thicker cast section could be produced without crystallising under the reported conditions.

Dataset overview

The supplied Excel workbook contains 594 alloy records and 14 columns: one alloy ID, one composition string, eleven numerical descriptors and Dmax (mm). Dmax ranges from 0.2 to 72 mm, with a median of 3.5 mm and a mean of 5.16 mm. The distribution is strongly right-skewed, meaning that most values are small while a few values are much larger.

Column	Presumed meaning	Range	Mean +/- SD
CalRad	Calculated atomic-radius descriptor	1.11 - 3.17	2.25 +/- 0.39
E	Electronegativity-type descriptor	1.15 - 1.87	1.46 +/- 0.16
Phi	Miedema work-function parameter	3.19 - 5.38	4.27 +/- 0.45
nWS	Miedema electron-density parameter	1.01 - 1.90	1.44 +/- 0.21
B	Bulk modulus (GPa)	29.2 - 223.2	111.6 +/- 48.9
Tm	Melting temperature (K)	959 - 2349	1554 +/- 350
Alpha	Thermal expansion coefficient	8.7 - 35.9	13.2 +/- 3.9
Hm	Enthalpy-type descriptor (kJ/mol)	8.2 - 28.0	15.3 +/- 5.1
MHc	Molar heat capacity (J/mol K)	20.6 - 32.2	24.7 +/- 1.5
Ro	Density-type descriptor (g/cm3)	1.94 - 13.31	6.33 +/- 1.77
k	Thermal conductivity (W/m K)	15.7 - 176.0	66.9 +/- 44.0
Dmax	Maximum critical casting diameter (mm)	0.2 - 72	5.16 +/- 5.99

Important: the Excel file does not include a data dictionary. The physical meanings shown above are inferred from the column names and value ranges, so the original source definitions should be checked before the report is treated as a definitive description of each descriptor.

10. Data checks and preparation

- **No missing numerical values.** All 594 records were retained for the descriptor-based PCA.
- **Missing composition strings.** 184 records (Alloy_ID 411-594) have no composition string. They remain in the PCA but are shown as "Unlabelled" in family-based plots.
- **Repeated alloy IDs.** There are 591 distinct IDs; IDs 10, 38 and 93 each appear twice. Row position was therefore used as the internal record key.
- **Repeated composition/descriptor groups.** Eight composition-and-descriptor groups have more than one reported Dmax value. Examples include Cu50Zr50 (25 mm vs 3 mm) and Pd77Cu6Si17 (2 mm vs 72 mm). There are no fully identical rows across all 14 columns. These differences are retained as part of the experimental scatter; the 72 mm Pd-Cu-Si value is excluded from the candidate list.
- **Composition normalisation.** Composition strings were parsed into atomic fractions and re-normalised to 100 at.%; five strings initially summed to values between 90% and 115%.
- **Dmax transformation.** log10(Dmax) is used for plot colouring and selected correlations because the raw Dmax distribution is strongly skewed.

PCA workflow

1. Select the eleven numerical descriptor columns and arrange them into a 594 x 11 data matrix. Alloy_ID and the composition text are not used as PCA input variables.
2. Standardise every descriptor using a z-score: subtract its mean and divide by its standard deviation. This places descriptors measured in different units on a comparable scale. Without standardisation, a variable such as melting temperature could dominate simply because its numerical values are large.
3. Apply PCA to the standardised matrix using singular value decomposition. The principal components are new, uncorrelated axes that capture decreasing amounts of variation in the dataset. The loading values describe how strongly each original descriptor is associated with a component.
4. Choose the number of useful components by looking at the Kaiser rule (eigenvalue greater than 1), the scree-plot elbow and cumulative explained variance (80% and 90% reference levels).
5. Keep Dmax as a supplementary variable. It is not allowed to influence the PCA axes; instead, its relationship with the component scores is calculated after PCA. This avoids using the performance measure to construct the map that is later used to screen alloys.
6. For 410 records with composition strings, parse elemental fractions and correlate them with the PC scores. Alloy family is assigned from the element with the highest atomic fraction, with La, Ce, Pr, Nd, Y, Gd, Dy, Er, Ho, Tm and Sc combined into the rare-earth (RE) family.

> **ROBUSTNESS CHECK**
>
> A second PCA run used log10(Dmax) as an active twelfth variable. PC1/PC2 changed from 66.5%/12.7% to 61.0%/13.3%, and Dmax loaded 0.03 on PC1 and 0.64 on PC2. The overall interpretation of the map remained similar.

11. Results and discussion

11.1 Correlation structure

Figure 8 shows the Spearman correlation between the eleven descriptors and log10(Dmax). A large group of descriptors changes together, while a smaller group tends to change in the opposite direction.

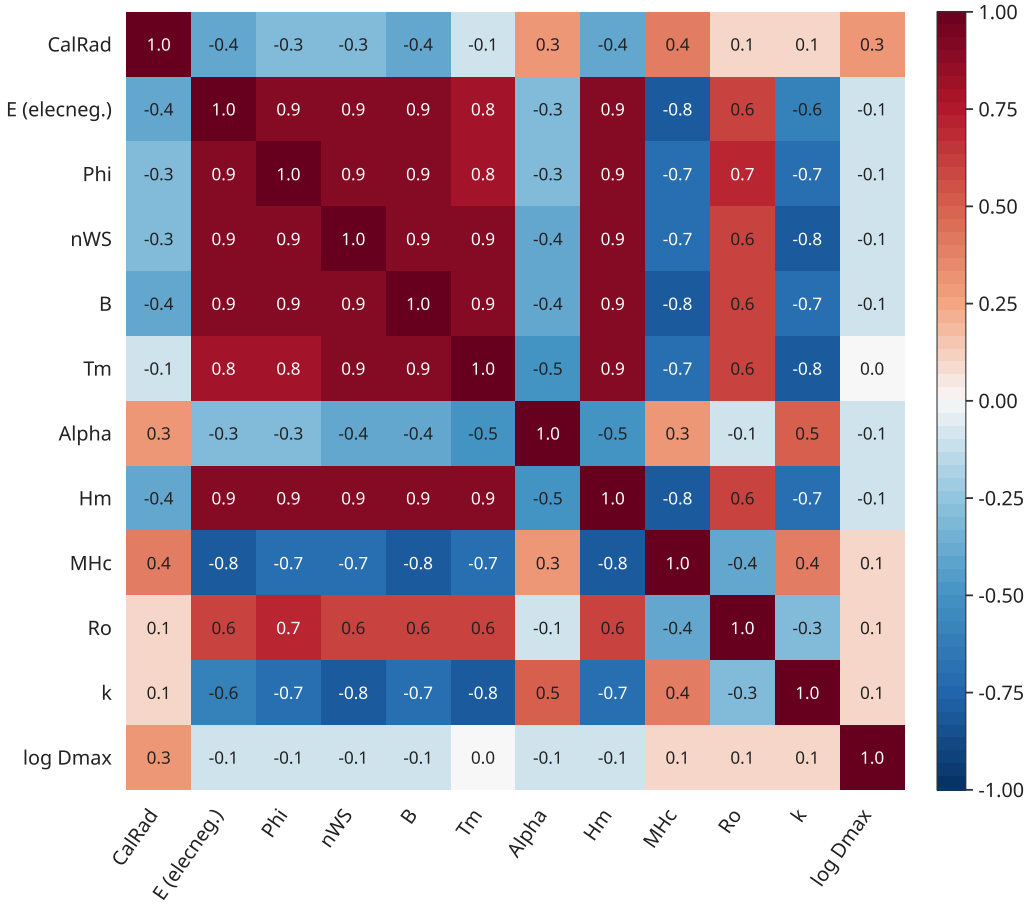


Figure 8. Spearman correlation matrix of the eleven descriptors and log10(Dmax).

- **Descriptors that move together.** nWS, B, Hm, Tm, E and Phi are strongly positively related (most pairwise correlations are above 0.8). These six columns therefore contain overlapping information, which helps explain why PCA can describe the dataset with only a few components.
- **Descriptors that tend to oppose them.** Thermal conductivity (k), molar heat capacity (MHc) and, to a lesser extent, thermal expansion (Alpha) show negative relationships with the main group.
- **Relationship with glass-forming ability.** log10(Dmax) has only weak correlation with any single descriptor (absolute correlation no greater than about 0.25). This suggests that Dmax is not controlled by one descriptor alone.

> **METALLURGICAL READING**
>
> BMG behaviour depends on several factors acting together - atomic size, chemical interactions, thermal properties and processing. The correlation matrix shows that some descriptors carry similar information, but it does not establish a cause-and-effect relationship.

11.2 How much variation do the components explain? (Q1)

The scree plot shows a sharp drop after PC1 and PC2. The first two components capture 79.2% of the total variation, while PC3 raises the total to 87.7%. Four components are needed to pass 90%.

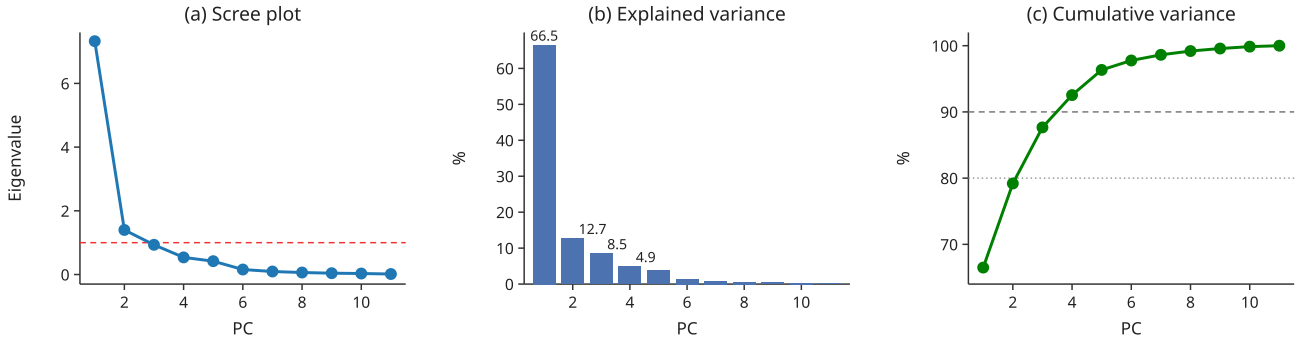


Figure 9. Scree plot, explained variance per component and cumulative explained variance.

Component	Eigenvalue	Variance (%)	Cumulative (%)	Interpretation
PC1	7.326	66.5	66.5	Dominant axis
PC2	1.399	12.7	79.2	Last component above the Kaiser limit
PC3	0.933	8.5	87.7	Just below 1; still useful for detail
PC4	0.538	4.9	92.6	Takes the total above 90%
PC5	0.419	3.8	96.4	Scree-plot tail begins

A two-dimensional map is therefore suitable for an initial view of alloy-family regions and broad descriptor trends. It should not be treated as a complete description: it leaves 20.8% of the total variation outside the PC1-PC2 plane. PC3 (8.5%) also helps separate some Pd-P and Mg-Cu-RE alloys that overlap in the first two components.

11.3 Which descriptors control each component? (Q2 and Q3)

A loading indicates how strongly an original descriptor is associated with a principal component. Large positive and negative values mean that the descriptor contributes strongly in opposite directions along that component.

Descriptor	PC1	PC2	PC3	PC4	Reading
CalRad	-0.38	+0.78	+0.17	+0.42	Main contributor to PC2
E	+0.93	-0.22	+0.15	+0.03	Part of the PC1 group
Phi	+0.92	+0.15	+0.21	-0.16	Part of the PC1 group
nWS	+0.98	+0.02	+0.05	+0.08	Largest PC1 loading
B	+0.97	-0.13	+0.02	0.00	Part of the PC1 group
Tm	+0.94	+0.15	+0.02	+0.12	Part of the PC1 group
Alpha	-0.42	-0.29	+0.83	+0.17	Main contributor to PC3
Hm	+0.96	-0.04	-0.03	+0.03	Part of the PC1 group
MHc	-0.75	+0.46	+0.06	-0.32	Opposes PC1; supports PC2
Ro	+0.67	+0.49	+0.32	-0.39	Contributes to PC1 and PC2
k	-0.76	-0.36	+0.21	-0.20	Opposes PC1

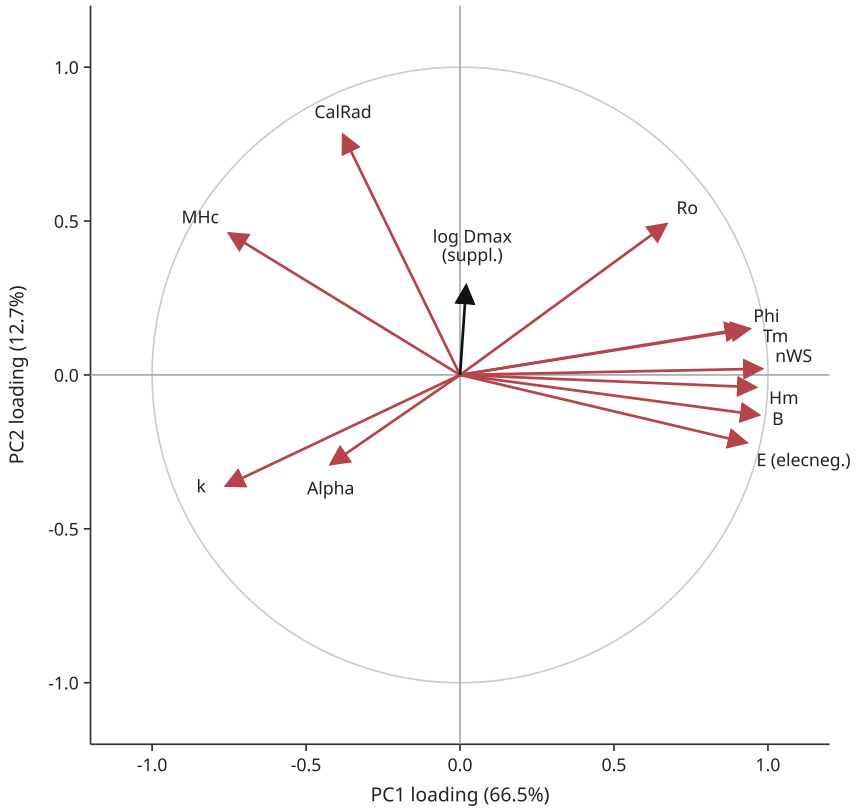


Figure 10. Loading plot for PC1-PC2 with the unit circle; log10(Dmax) is a supplementary vector.

PC1 (66.5%) can be read as a combined descriptor trend associated with bonding-related parameters and higher stiffness/melting behaviour. nWS, B, Hm, Tm, E and Phi all load strongly in the positive direction (+0.92 to +0.98). Fe-, Co-, Ni- and Pd-rich alloys tend to lie on this side of the map, while Ca-, Mg- and RE-rich alloys tend to lie on the negative side. The variables k and MHc point in the opposite direction.

PC2 (12.7%) is mainly associated with the calculated atomic-size descriptor CalRad (+0.78), with additional contributions from Ro (+0.49) and MHc (+0.46). It can be interpreted as an atomic-size/density-related direction. PC3 (8.5%) is mainly associated with thermal expansion Alpha (+0.83). These labels are interpretations of the descriptor pattern, not direct measurements of a single physical mechanism.

The supplementary log10(Dmax) vector has a weak relationship with PC1 (Pearson r = +0.02) and a stronger, though still moderate, relationship with PC2 (r = +0.29). In this dataset, Dmax is therefore more aligned with the PC2 direction than with the main PC1 descriptor trend.

11.4 Element-level associations (Q2)

To interpret the principal components in terms of alloy chemistry, the atomic fraction of each element was correlated with the PC scores for the 410 alloys that have composition strings. Positive values mean that a higher fraction of an element tends to occur at higher scores on that component; negative values indicate the opposite trend.

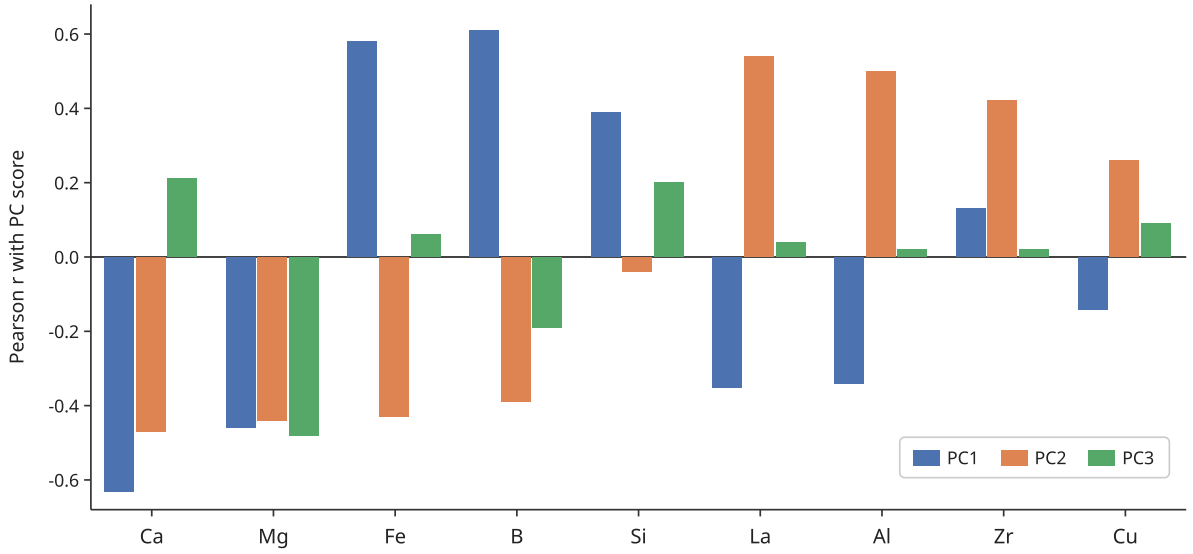


Figure 11. Pearson correlation of atomic fraction (410 labelled alloys) with PC1-PC3 scores for selected elements.

- **PC1 - transition-metal/metalloid side.** B (+0.61), Fe (+0.58) and Si (+0.39) are positively associated with PC1. Ca (-0.63), Mg (-0.46) and Zn (-0.41) are negatively associated. This separates many metalloid-containing late-transition-metal glasses from simple-metal-rich compositions.
- **PC2 - atomic-size and alloy-family trend.** La (+0.54), Al (+0.50) and Zr (+0.42) show positive associations. Ca (-0.47), Mg (-0.44), Fe (-0.43) and B (-0.39) show negative associations.
- **PC3.** The strongest selected negative association is with Mg (r = -0.48).

A separate PCA using the composition matrix directly (26 elements present in at least 10 alloys) explained only 14.5%, 8.8% and 8.3% in its first three components. The composition matrix is sparse and high-dimensional; the composition-derived descriptors provide a more compact map for this dataset.

11.5 Do BMG families form groups? (Q4)

The PC1-PC2 score plot places alloys with similar descriptor patterns near one another. The groups are not perfectly separated, but several broad family regions can be recognised.

Figure 12 – PC1-PC2 score plot by family

Insert the original figure here. This plot shows the individual PCA scores of each alloy record, which can only be drawn from the BMG_PCA.xlsx data (not supplied with the reference reports).

Figure 12. PC1-PC2 score plot coloured by dominant-element family; the 184 records without composition strings are shown in grey.

Ca-based alloys lie far to the left (mean PC1 about -4.7). Mg-based alloys sit left of centre (about -2.4, -1.0), while RE-based alloys form an upper-left region (about -2.4, +1.8). Zr/Cu-based alloys occupy a compact upper-central region (about +0.3, +1.0). Pd-based alloys lie towards the right-centre, and Fe- and Co-based alloys extend to the far right (mean PC1 about +4.0 and +5.0, respectively).

The silhouette score is 0.30, which indicates moderate separation rather than completely distinct clusters. A 5-nearest-neighbour classifier recovers the family label with 84% cross-validated accuracy using five folds; using three PCs also gives 84%. Zr- and Cu-based alloys overlap strongly. This is reasonable because many are variants of the same Cu-Zr-(Al, Ag, Ti) glass-forming systems and can receive different labels simply because a different element is dominant.

Among the 184 unlabelled records, 39 fall in the Zr/Cu region, 36 in the Fe/Co region, 6 in the Ca region and 4 in the RE region. These possible family assignments are based only on PCA position; the missing composition strings must be recovered before they can be confirmed.

11.6 Critical casting diameter and missing performance targets (Q5)

Only 84 of the 594 alloys (14.1%) have Dmax >= 10 mm. Their positions are not spread evenly across the score plot: many occur near the upper-central Zr-Cu region, with another concentration in the La-rich region.

Figure 13 – PC1-PC2 score plot coloured by log10(Dmax)

Insert the original figure here. This plot shows the individual PCA scores of each alloy record, which can only be drawn from the BMG_PCA.xlsx data (not supplied with the reference reports).

Figure 13. PC1-PC2 score plot coloured by log10(Dmax).

The median high-Dmax alloy lies near PC1 = +0.2 and PC2 = +1.0. The Spearman correlation between Dmax and PC2 is +0.29 (p about 10^-12), compared with -0.07 for PC1 (p = 0.08) and -0.10 for PC3 (p = 0.02). Across PC2 quartiles, median Dmax rises from 2 to 3 to 4 to 5 mm, while the share of alloys with Dmax >= 10 mm rises from about 7% and 5% to 18% and 26%.

Score-plot region	n	Median Dmax (mm)	Share with Dmax >= 10 mm	Mean B (GPa)
Zr-Cu zone: PC1 -0.3 to 0.9; PC2 0.5 to 1.7	113	6.0	36%	-
RE (La) zone: PC1 < -1.8; PC2 > 1.2	60	5.0	28%	-
Fe/Co zone: PC1 > 3	103	2.5	4% (4 alloys)	193
Ca zone: PC1 < -3.5	55	2.0	11% (6 alloys)	46
All alloys	594	3.5	14%	112

The Zr-Cu region contains about 2.6 times the overall share of alloys with Dmax >= 10 mm (36% versus 14%). The La-rich region contains about twice the overall share (28%). The Ca and Fe/Co regions have lower median Dmax values, showing that a very high or very low stiffness/melting-related descriptor level does not by itself produce a large glassy casting section.

> **IMPORTANT LIMITATION ON Q5**
>
> The workbook contains no Tg, Tx, Delta Tx (Tx - Tg), yield strength, tensile strength or hardness column. Therefore, this analysis cannot identify alloys with a wide supercooled liquid region or high strength directly. Bulk modulus B is used only as a stiffness-related proxy: the Fe/Co region has mean B about 193 GPa versus 112 GPa overall. Bulk modulus measures resistance to volume compression; it is not a direct strength measurement. The strength interpretation remains a hypothesis to test.

11.7 Reading the biplot and selecting candidates (Q6)

The biplot places the alloy scores and descriptor directions on the same map. The arrows show the directions in which each descriptor increases; points lying in a similar direction tend to have higher values for that descriptor.

Figure 14 – Biplot with candidates C1-C7

Insert the original figure here. This plot shows the individual PCA scores of each alloy record, which can only be drawn from the BMG_PCA.xlsx data (not supplied with the reference reports).

Figure 14. Biplot: alloy scores coloured by log10(Dmax), descriptor vectors shown in red, and candidate alloys C1-C7 circled.

The Zr-Cu cluster lies just above the origin, in the direction of CalRad and Ro, and is not strongly aligned with either the high-stiffness vectors on the right or the soft-metal vectors on the left. The La-rich cluster lies towards the upper-left, along the MHc and CalRad directions. Fe/Co alloys extend along the E-B-Hm-nWS direction.

The shortlist uses three types of evidence: (i) the candidate position in score/biplot space, (ii) the reported Dmax value, and (iii) the average Dmax of the eight nearest neighbours in PC1-PC3 space, referred to as the neighbourhood mean. High-Dmax anchors are included to represent productive regions. Lower-Dmax compositions surrounded by higher-Dmax neighbours are included as under-explored candidates. These are hypotheses for further work, not predictions of the result of casting.

12. Candidate alloys for further investigation

Tag	Alloy (ID)	Dmax (mm)	Nbr. mean (mm)	PC1 / PC2	Role	Reason for selection
C1	Zr48Cu34Pd2Ag8Al8 (288)	25	14.8	0.22 / 1.11	Anchor	Located near the centre of the high-Dmax Zr-Cu zone; a useful reference composition.
C2	Cu34Zr50Ag8Al8 (60)	20	13.2	0.17 / 1.12	Anchor	Zr-rich end of the same Cu-Zr-Ag-Al group.
C3	Cu38Zr46Ag8Al8 (58)	3	11.6	0.20 / 0.95	Under-explored	Sits near higher-Dmax neighbours; the 4 at.% Cu/Zr shift may be associated with a strong Dmax change.
C4	Cu45Zr45Ag7Al3 (51)	3	11.3	0.38 / 1.01	Under-explored	Cu-rich and Al-lean member of the same region; reported Dmax is below its neighbours.
C5	La62Al14Cu11.66Ag2.34Ni5Co5 (187)	16	8.5	-2.42 / 1.94	Anchor	A high-Dmax member of the La-rich region and a second glass-forming family.
C6	La62Cu12Ni12Al14 (177)	3	9.6	-2.32 / 1.94	Under-explored	Same La-rich region as C5; an Ag-free composition worth testing.
C7	Fe61B15Mo7Zr8Co7Y2 (305)	16	3.7	3.45 / -0.18	Stiffness-oriented outlier	High-B region (B about 188 GPa); isolated, so it should be verified before further claims.

Candidate Dmax values were checked against the supplied workbook: C1 (ID 288) = 25 mm and C2 (ID 60) = 20 mm. The remaining candidate values are also matched to their corresponding alloy records.

The neighbourhood means are carried over from the original report to preserve the reported analysis. The supplied notebook does not contain the code used to calculate them; a separate reproducibility note in Section 14 explains this limitation.

Compositions excluded from the shortlist

Pd77Cu6Si17 is excluded because the dataset contains the same composition and descriptor values with Dmax values of 2 mm and 72 mm. The 72 mm value is treated as an outlier until the source can be checked. Co50Cr15Mo14C15B6 (25 mm) is also excluded because it is an isolated Co-based outlier; C7 is a better documented strength/stiffness-oriented candidate. Records without composition strings cannot be recommended, although several records (IDs 565-587) appear in the Zr-Cu region and should be identified first.

12.1 Suggested validation plan

- **Data-driven modelling.** Train a random forest or gradient-boosted model to predict log(Dmax), using family-grouped cross-validation. Build a classifier for Dmax >= 10 mm and use SHAP values to

inspect whether model importance agrees with the PCA loadings. Gaussian-process or Bayesian optimisation can be explored around C1-C4.

- **Clustering.** Apply k-means, Gaussian-mixture and hierarchical clustering to PC1-PC3. Use t-SNE or UMAP as a non-linear comparison to check whether the family regions remain stable.
- **Thermodynamic and atomic-scale screening.** Use CALPHAD to estimate liquidus temperature, deep-eutectic location and Trg = Tg/Tl. Check mixing enthalpy and atomic-size mismatch (Inoue-type criteria) for the Zr-Cu-Ag-Al and La-based candidates.
- **Casting and phase confirmation for C1-C6.** Prepare arc-melted alloys and suction- or injection-cast wedge/rod samples at 3, 5, 10, 15 and 20 mm. Use XRD to check for an amorphous halo and DSC to measure Tg, Tx and Tl, then calculate Delta Tx and Trg. Repeat C3 and C4 using fixed purity and cooling conditions to check whether their low Dmax values are repeatable.
- **Mechanical testing for C7.** Repeat casting and DSC, then carry out compression tests for yield strength and plastic strain, together with nanoindentation.

13. Limitations of PCA for alloy design (Q7)

- **Linear and unsupervised.** PCA identifies directions with the largest variation in the descriptor matrix; it does not find directions that necessarily explain Dmax. Here, PC1 explains 66.5% of the variation but has almost no correlation with Dmax (r about +0.02 for log10(Dmax)). PCA is therefore a screening and visualisation tool, not a Dmax prediction model.
- **Information loss.** The PC1-PC2 view leaves 20.8% of the total variation outside the map. A useful Dmax-related pattern could also occur in a lower-variance component such as PC3 or PC4.
- **Composition-derived averages.** Different compositions can have similar descriptor values. The dataset contains repeated descriptor groups, and these averaged descriptors do not capture cooling rate, melt purity, oxygen content, casting method or melt fluxing - all of which can affect Dmax.
- **Noisy and skewed target.** Dmax depends on the experimental limits and conditions reported in different studies. Conflicting entries such as 3 vs 25 mm for Cu50Zr50 and 2 vs 72 mm for Pd77Cu6Si17 illustrate this issue. The values are reported maxima, not averages.
- **Incomplete and imbalanced family labels.** Family sizes range from only a few records to several dozen, and 184 records have no composition string. Literature datasets may also over-represent successful glasses because unsuccessful compositions are less likely to be reported.
- **Redundant inputs and interpretation.** Six descriptors have PC1 loadings above 0.92 in magnitude, so the main component contains several strongly related inputs. A principal component is a combination of descriptors; a high PC2 score is not, by itself, an alloy recipe.
- **Missing target properties.** The workbook has no Delta Tx, Tg, Tx, yield strength, tensile strength or hardness values. These questions can only be discussed using the limited Dmax analysis and the stiffness-related B proxy.

These limitations support the proposed next steps: family-aware supervised modelling, checks on clustering stability, thermodynamic screening and, ultimately, controlled casting followed by XRD, DSC and mechanical testing.

Conclusions

The first two principal components explain 79.2% of the descriptor variation, rising to 87.7% with PC3. PC1 mainly represents the shared trend across nWS, B, Hm, Tm, E and Phi, opposed by k and MHc. PC2 is mainly associated with CalRad, Ro and MHc, while PC3 is strongly associated with Alpha.

The score plots show recognisable BMG family regions, with substantial overlap between Zr- and Cu-based alloys. High Dmax values are most concentrated in the Zr-Cu-(Ag, Al) region and the La-rich region, and Dmax is more closely associated with PC2 than PC1. Seven compositions form a shortlist for further investigation, including reference anchors, under-explored neighbours and one stiffness-oriented Fe-based outlier.

Overall, PCA narrows the search space and helps form testable hypotheses. It cannot identify a universally optimal BMG on its own. Experimental casting, phase confirmation, thermal analysis and mechanical testing - supported by supervised models - are needed before any shortlisted composition can be considered a successful alloy.

14. Reproducibility and verification notes

The supplied Excel file was checked for row count, missing values, descriptor ranges and the candidate records. The PCA variance fractions and leading loadings in the report are consistent with a fresh PCA of the 594 x 11 descriptor matrix after z-score standardisation.

- **PCA values checked.** PC1 = 66.49%, PC2 = 12.70%, PC3 = 8.47%, PC4 = 4.88% and PC5 = 3.80%; these agree with the rounded values in the report.
- **Candidate Dmax checked.** The Excel file confirms C1 (ID 288) = 25 mm, C2 (ID 60) = 20 mm, C3 = 3 mm, C4 = 3 mm, C5 = 16 mm, C6 = 3 mm and C7 = 16 mm. The shortlist now places each value beside the correct alloy.
- **Notebook coverage.** The uploaded notebook reproduces dataset loading, standardisation, base PCA, explained variance and basic score/biplot visualisations. It does not include the code for all extended analyses in this report, including the Spearman descriptor matrix, element-to-PC correlations, family labelling, silhouette/5-nearest-neighbour validation, regional Dmax statistics or the candidate neighbour means. The complete analysis cells or scripts should be attached if full computational reproducibility is required.
- **Neighbourhood means.** The original neighbourhood means (14.8, 13.2, 11.6, 11.3, 8.5, 9.6 and 3.7 mm for C1-C7) are retained in the candidate table to preserve the original reported results. A fresh Euclidean 8-neighbour calculation using the supplied PCA scores in raw PC1-PC3 coordinates gives different values (16.2, 14.4, 13.4, 14.8, 9.4, 11.4 and 3.9 mm). Because the supplied notebook does not document the original neighbour calculation, confirm the intended distance/scaling method before relying on these means.
- **Descriptor definitions and sources.** The descriptor labels are interpreted from the workbook headers and ranges because a data dictionary was not supplied. The source publication for the dataset and authoritative definitions for the descriptors should be added to the final bibliography when available.